IE4728 Enterprise Network Design Project

Design Project Briefing (Week 6 and 7)

Tutorial Group	F52	F53
Venue	TR120	TR120
Time	0930-1230	1430-1730

Students form Design Groups of no more than 3 students each

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IE4728 Enterprise Network Design Project

Design Project Implementation (Week 8 onwards)

Group Schedule	Design Project	Network Lab
Week 8 - 10	Group A	Group B
Week 11 - 13	Group B	Group A

Design Group size: max 3 students per group
Lab Group size: max 5 students per group

Project venue (Information System) at S2-B3b-06
Lab venue (Communication Lab II) at S2-B4-17

IE4728 Course Assessment

- Examination: 50%

  Cover contents in

  - ✓ Lecture,
  - ✓ Project Briefing and
  - ✓ Laboratory sessions

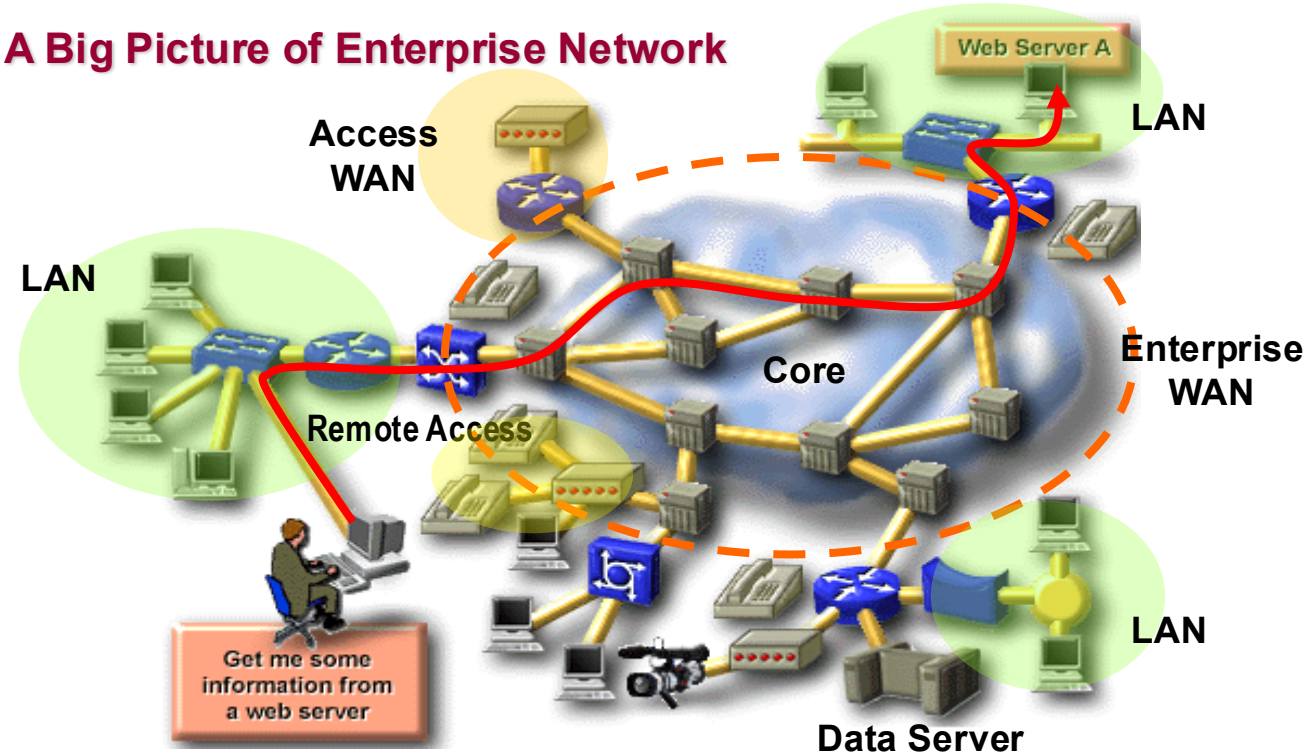
- Design Project: 44%
  - ✓ Project report
  - ✓ Design simulation
  - ✓ Power Point slides
  - ✓ Include weekly progress assessment

- Laboratory Exercises: 6%

Design Project Overview

- Project Description
- Design Requirements
- Project Submission

A Big Picture of Enterprise Network



An enterprise network consists of a group of LANs interconnected using WANs, and is under the control of one big organization.

Enterprise network includes:

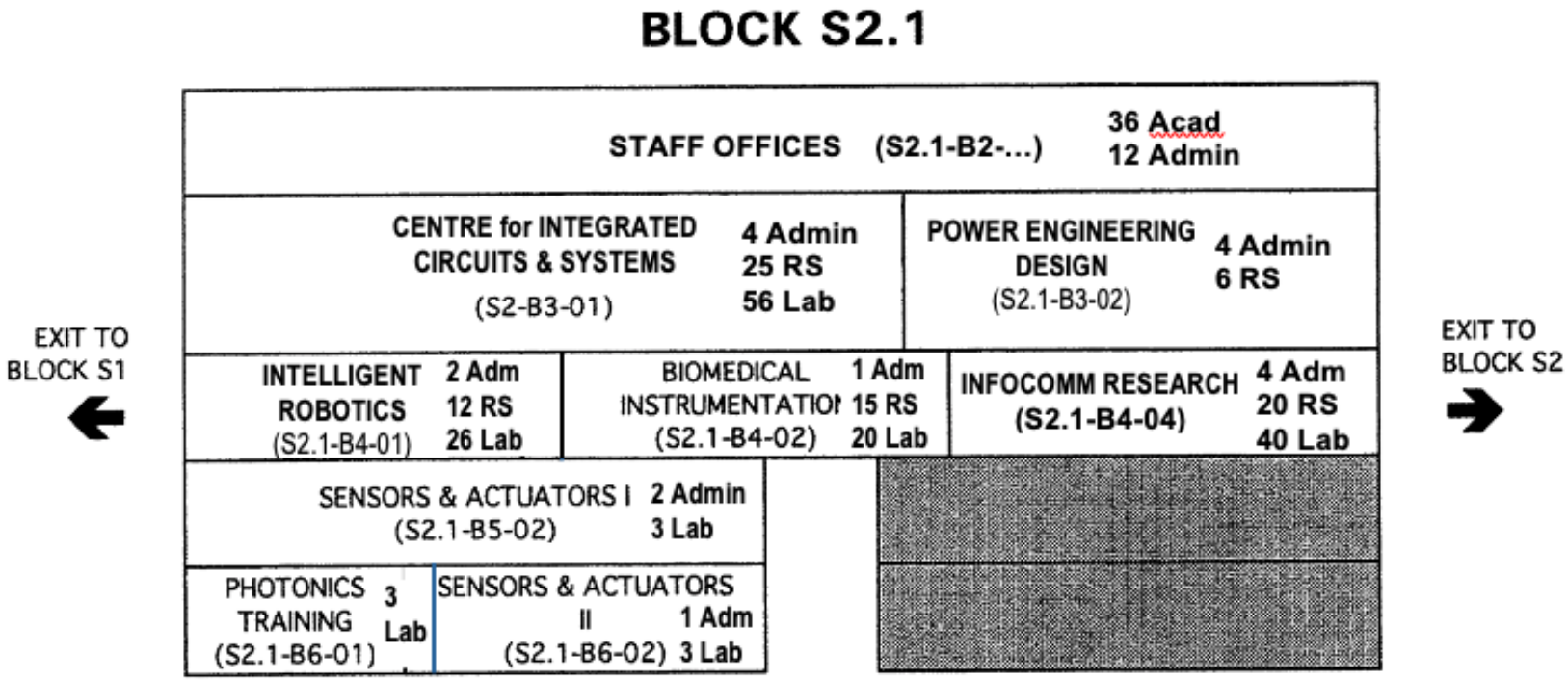
- **LANs**: physically connect users in a building or group of buildings with relatively high bandwidth for sharing devices on local networks
- **WANs**: logically interconnect LANs to provide access to computers or file servers in other locations
- **Internetworking devices and protocols**: Devices (e.g., switches, routers, gateways) and Protocols (e.g. ARP, IP, TCP/UDP, HTTP/DNS)

Aim of Project

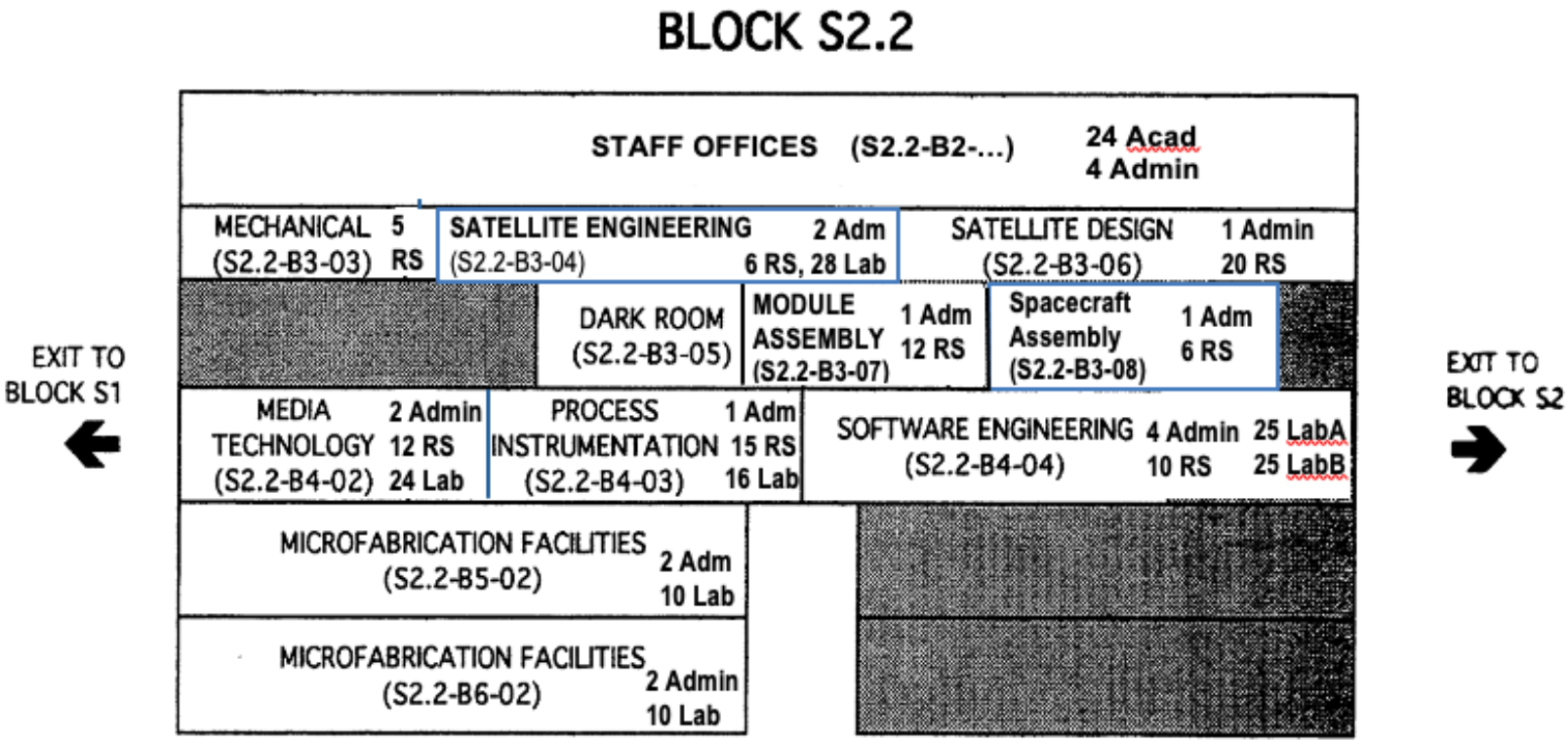
To design an enterprise network to facilitate:

- Interworking of workgroups.
- Resource sharing.
- A case study on 2 completed facilities (Block S2.1 and S2.2) in the School of EEE.

Distribution of Hosts and User Groups in S2.1



Distribution of Hosts and User Groups in S2.2



Design Requirements

- **Network Requirements**
  - Interworking of workgroups
  - Network resource sharing (file/data, servers, printers)
- **User Requirements**
  - 4 Workgroup: Academic (Acad), Administrative (Admin), Research students (RS), and lab users (Lab)
  - Users collaborate within same workgroup and across different network segments
  - Lab groups do not interoperate nor share resources with other groups
- **Network Design features:**
  - **LAN segmentation** (and augmentation) of physical networks
  - **VLSM subnetting** (and supernetting) of logical networks
  - **Optimized** layered design for performance and robustness
- **Access control lists for:**
  - Host and network access
  - Interworking of workgroups
  - Internetworking and resource sharing

Design Work

- **Design approach**
  - Top-down analysis of design requirements
  - Bottom-up implementation of layered network design
  - Five (5) design templates to guide design and implementation
- **Network design model**
  - Modular and hierarchical layered design: core, distribution, and access layers
- **Design implementation**
  - Minimize address wastage while maximizing route aggregation.
  - Optimized load distribution and robustness against node/link failures
- **Access control policies**
  - ACLs for internetworking of subnets and interworking of workgroups
- **Network design simulations**
  - Simulation of design and implementation using Packet Tracer

Project Deliverables

Design report

- 7-chapter report incorporating the five design table templates
  - Comprises chapters on Introduction & Objectives, Top-down analysis, Bottom-up implementation, Multilayered network design, Network access control, Simulation results, and Conclusion
  - All design features should be appropriately justified and substantiated
    - No right or wrong designs, only good and bad ones
  - Include a list of group members' contributions

Design simulation

- Simulate the network design using Packet Tracer
  - Produce simulation results of the enterprise network and all design requirements as listed in the project description.
  - Include in the .pkt simulation file, all relevant scenarios on packet routing and access control.
  - Ensure all default program settings are correctly saved in the simulation file.

Design presentation slides

- Present an executive summary of the design approach and key design features in Power-Point slides

Design Project Submission

All design project submissions are to be made on the course site

- **Design report**
  - Name the report in the format: "Fxx-Gyy_Title"
    - e.g. F51-G02_Report, F51-G02_Simulation
  - 1 softcopy report to be submitted to Turnitin
- **Design simulation***
  - 1 softcopy of the design simulation results generated using the Packet Tracer v8 software.
- **Design slides***
  - 1 softcopy of the executive summary in power-point slides

  * zip into one file and submit as an assignment on the course site
- **Submission deadline**
  - one week after the 3-week design session (i.e. the 4th week)

Project Implementation Schedule

Reasonable weekly progress (8th week onwards):

- Week 1: Completion of Table 1
  - Top-down analysis of network requirements
  - Completion of LAN segmentation of physical networks
  - Completion of subnetting and superneting of logical networks
- Week 2: Completion of Table 2 and 3
  - Bottom-up network implementation of logical networks
  - Completion of Host IP assignment and server placement
- Week 3: Completion of Table 4 and 5
  - Access control lists (ACL) for network security and data sharing
  - Completion of the design and placement of ACLs
  - Created simulation scenarios to test the network design

Table 1a: Template for Network Address Assignments

Template for Network Address Assignments using VLSM Subnetting								
Location	Acad		Admin		RS		Lab	
Block S2.1	Hosts	Net#/Subnet Bits	Hosts	Net#/Subnet Bits	Hosts	Net#/Subnet Bits	Hosts	Net#/Subnet Bits
S2.1-b2	36	x.x.28.01/26	12					
s2.1-b3-01			4		25		56	x.x.30.01/26
s2.1-b3-02			4		6			
s2.1-b4-01			2		12		26	
s2.1-b4-02			1		15		20	
s2.1-b4-04			4		20		40	
s2.1-b5-02			2				3	x.x.30.0010/28
s2.1-b6-01							3	
s2.1-b6-02			1				3	x.x.30.0010/28
Total hosts*	**36+n**		**30**		**78**		**151**	
Block S2.2								
sS.2-b2	24	x.x.28.001/27	4					
s2.2-b3-03					5			
s2.2-b3-04			2		6		28	
s2.2-b3-06			1		20			
s2.2-b3-05								
s2.2-b3-07			1		12			
s2.2-b3-08			1		6			
s2.2-b4-02			2		12		24	
s2.2-b4-03			1		15		16	
s2.2-b4-04			4		10		2x25	
s2.2-b5-02			2				10	
s2.2-b6-02			2				10	
Total hosts*	**24+n**		**20**		**86**		**138**	
* to include additional n addresses for printers, servers, and router interfaces.								

Table 1b: Template for Network Address Summarization

Supernetting of Subnets for Network Address Summarization			
Network	Net#/Supernet Bits	Subnets	Description of logical group assignments
x.x.28.0	x.x.28.0/25	x.x.28.01/26 x.x.28.001/27	Supernet for aggregration of Acad groups in S2.1 and S2.2
			<Admin groups>
			<RS groups in S2.1>
			<RS groups in S2.2>
x.x.30.0			Network for all Lab groups in S2.1
x.x.30.32/28			Subnet for combined Lab groups in S2.1-b5-02 & S2.1-b6-02
			<All Lab groups in S2.2>
			<Combined Lab groups in S2.2>

Table 2: Template for Host IP Address Assignments

User Group	Locations	No of Hosts	No. of Printer/ Server	Sub Network Address	Sub-net Mask	Router Address	First Host Address	Last Host Address	Printer Address	Server Address	No. used / unused
Lab_2.1	CICS	56	1 / 1	207.21.30.0	/26	x.x.x.65	x.x.x 66	x.y.z 121	x.x.x.122	x.x.x.123	59 / 3
	Intelligent Robots	26									
	Biomedical	20									
	Infocommunication Research	40									
	Sensors & Actuators i	3	1/1		2x /29 or /28 ?						2x 6/0 or 12/2 ?
	Sensors & Actuators II	3	1/0								
	Photonics	3									

Table 2. Table template for host address assignments

Tables 3: Templates for Server Address Assignment and Placement

Server Name	Server IP Address	Subnet Mask	Attached Router Name	Router Interface
HTTP server				
DNS server				
SQL Server				
Acad Server				
Admin Server				
RS Server				
Email Server				
(all ntwk devices)				
Etc…				

Table3. Sample template for server address assignments

Different type of servers:

- Local - labs
- Group - workgroups
- Enterprise – all groups
- Server farm – group of servers

Tables 4: Template for Assignment and Placement of Access Control Lists

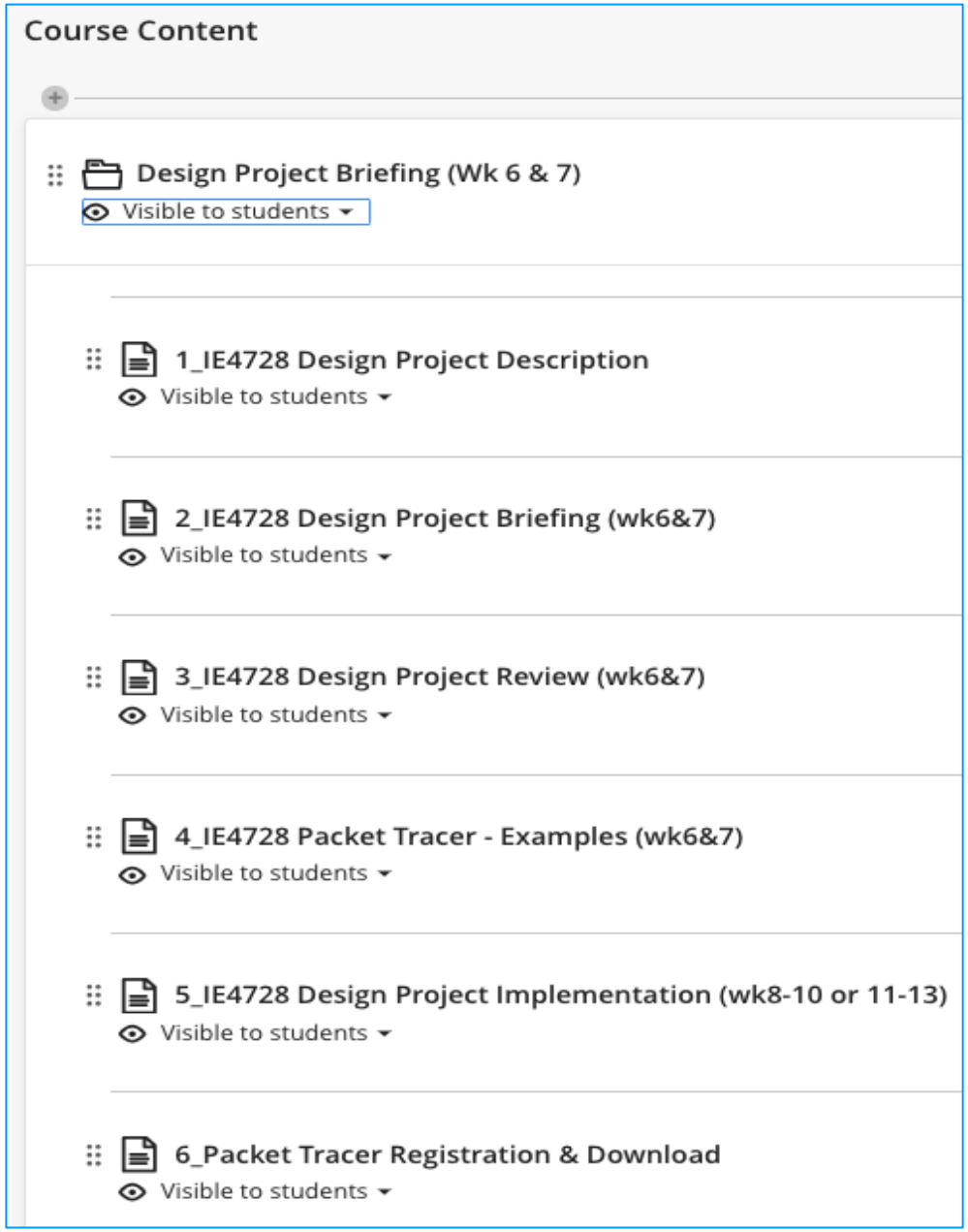
Network Name	Router Name	Router Interface	Network #	Subnet mask	IP Address	Access List Number
RS Network	RS S2.1	0/0				x
		1/0				x
		2/0				1 (OUT)
	RS S2.2	0/0				x
		1/0				x
		2/0				x
	RS	0/0				x
		1/0				123 (IN)
		2/0				
		3/0				

Table 4. Sample template for ACL assignments

Table 5: Template for Access Control List Statements

Network	Router Name	Router Interface	IP Address	Type	Access-List
RS Network	RS S2.2	1/0		In/Out	Extended IP access list 101 **// permit FTP service from RS to Acad** permit tcp x.x.x.x filter y.y.y.y filter eq 21 **// deny FTP service from RS to others** deny tcp .. .. .. any eq 21 **// permit the rest access** permit ip any any
	RS	0/0		In/Out	Extended IP access list 102 **// permit FTP service from Acad to RS** … **// deny FTP service from others to RS** … **// permit the rest access** …

Table 5. Sample template for access-control list statements


Course Content
Design Project Briefing (Wk 6 & 7)
Visible to students
1_IE4728 Design Project Description
Visible to students
2_IE4728 Design Project Briefing (wk6&7)
Visible to students
3_IE4728 Design Project Review (wk6&7)
Visible to students
4_IE4728 Packet Tracer - Examples (wk6&7)
Visible to students
5_IE4728 Design Project Implementation (wk8-10 or 11-13)
Visible to students
6_Packet Tracer Registration & Download
Visible to students